

Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis

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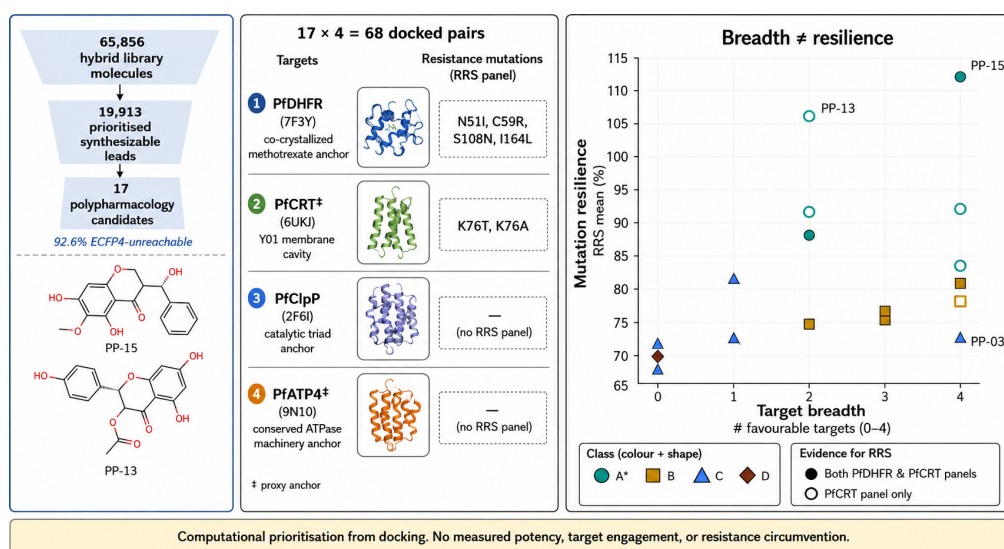
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Abstract

Can target breadth and mutation resilience be evaluated as distinct, complementary properties when prioritizing antimalarial chemotypes from African natural-product-inspired chemical space? We addressed this question by linking chemical-space expansion to target-specific docking and a per-target resistance-resilience score (RRS), while keeping computational evidence separate from biological activity. A hybrid library of 65 856 molecules, generated from 396 African natural products and 454 synthetic antimalarial compounds, yielded 19 913 computationally prioritized synthesizable leads. A 17-member polypharmacology-oriented cohort was evaluated by AutoDock Vina against PfDHFR, PfCRT, PfClpP, and PfATP4 using target-specific structural anchors. All 68 candidate–target pairs passed the geometric pose-quality criterion, with scores ranging from -12.01 – -4.63 kcal mol⁻¹. Per-target RRS was calculated from separate PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels, excluding non-binding wild-type targets from the denominator; PfCRT was re-docked against a corrected 3D7-like wild-type receptor (LYS-76) under a cavity-anchored grid, and a pipeline-null control showed that the corrected PfCRT channel carries no detectable K76T/K76A signal for this panel (RRS 99.1–100.6 %, null range 99.4–100.5 %). The cohort comprised seven A*, four B, and six C RRS profiles, spanning 73.2–115.6 %. After multiplicity correction, PNS correlated negatively with RRS ($\rho = -0.714$), RRS correlated positively with wild-type score magnitude ($\rho = 0.691$), and the number of favorable targets correlated positively with RRS ($\rho = 0.714$); ACSI–RRS was not significant ($\rho = -0.190$). The resulting target profiles define testable hypotheses about chemotype breadth and mutation resilience; they do not establish potency, target engagement, or resistance circumvention.



Keywords: African natural products; antimalarial discovery; polypharmacology; resistance resilience; molecular docking; chemical space; PfDHFR; PfCRT

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1 Introduction

Antimalarial resistance is a systems problem [1–3]. A compound that depends on one protein–ligand interaction may lose activity after a single parasite mutation [4], whereas a chemotype that engages several resistance-relevant processes could provide a broader mechanistic starting point [5–8]. This rationale does not imply that every computationally predicted multi-target profile is biologically useful: target breadth, mutation tolerance, physicochemical tractability, and experimental activity are distinct properties that must be measured separately.

African natural products provide a chemically rich but incompletely explored starting point for such studies [9–12]. Their ring systems and stereochemical complexity can complement synthetic antimalarial chemistry, while computational expansion can generate tractable analogues around privileged scaffolds. Yet a computational accessibility gap persists: target-specific docking campaigns in malaria drug discovery typically demand 500–1 000 CPU-hours per target–ligand combination [13, 14], placing systematic multi-target and mutation-panel screening beyond reach for many research groups in endemic regions where 94 % of malaria cases occur [1]. Centroid-based library reduction and automated target-anchored workflows can reduce this barrier by orders of magnitude while retaining structural diversity, provided that the resulting computational evidence is interpreted with appropriate caution. The challenge is to connect this chemical-space exploration to a target-level analysis without allowing the prioritization score to become a surrogate for activity.

We therefore assembled an integrated workflow with three separated layers. First, a chemical-space funnel combines African natural products and synthetic antimalarials, expands the library, and prioritizes synthesizable candidates. Second, target-anchored docking evaluates a locked polypharmacology-oriented cohort against four proteins with unequal structural evidence. Third, a mutation-aware RRS summarizes changes in docking estimates for declared computational PfDHFR and PfCRT mutant panels. Polypharmacology is read from the four-target docking profile; RRS is read from target-specific mutant comparisons. The two layers are joined by canonical molecular identity for exploratory interpretation, not by assuming that multi-target docking proves mutation resilience. Structural evidence is heterogeneous across the panel: PfDHFR carries a co-crystallized inhibitor, PfCRT is represented by a proxy cavity, and PfClpP and PfATP4 by mechanism-motivated regions without co-crystallized small-molecule ligands. Target breadth is therefore read as a pattern of target-specific evidence across a gradient of structural confidence, not as four equivalent demonstrations of biological engagement.

We asked whether this separation could produce a coherent map from chemical-space novelty to two experimentally testable properties: target breadth and mutation resilience. The analysis was designed around three distinctions: candidate sets remained separate, non-binding wild-type targets were excluded from RRS denominators, and docking-derived evidence was not treated as a surrogate for biological activity.

2 Materials and Methods

2.1 Chemical-space funnel

The upstream library combined 396 African natural products with 454 synthetic antimalarial compounds. The dataset was expanded via two complementary approaches: the Cheese API [15] performed similarity searches (Tanimoto 0.7, Morgan fingerprints radius 2, 2 048 bits) against ZINC15/ENAMEINE-REAL, retrieving 100 analogues per seed; and STONED-SELFIES [16] generated 100 variants per seed via single-character SELFIES mutations (alphabet size 50). Generated molecules were PAINS-filtered [17], validated via RDKit and Open Babel [18], and deduplicated by canonical SMILES, yielding 65 856 unique molecules (94.1 % Lipinski-compliant). A SMILES variational autoencoder [19] was used to obtain latent representations. K-means

clustering of the selected 64-dimensional representation yielded 484 centroids. This centroid-based sampling reduces computational cost from exhaustive docking (500–1 000 CPU-hours per target for full libraries) to tractable multi-target screening while preserving structural diversity through cluster representatives. Multi-parameter optimization combined docking-derived scores, DiffDock confidence [20], QED [21], an ADMET composite, and a Lipinski penalty [22]. The 76 centroids above the optimization threshold mapped to 53 clusters and expanded to 40 481 molecules; 19 913 molecules remained after synthetic-accessibility and MPO filtering.

Novelty was evaluated with Morgan fingerprints [23] (radius 2, 2 048 bits). In a 5 000-molecule sample, 92.6 % had maximum whole-molecule Tanimoto similarity below 0.4 to the African-NP seed set, while scaffold recovery was 69.3 % (70/101 annotated seed scaffolds). The candidate sets were locked before the target-anchored docking analysis.

2.2 Candidate-set identity and Set C selection

Set A comprised the MPO top-20 candidates, Set B comprised the MD top-20 candidates, and Set C comprised the 17-member polypharmacology-oriented cohort. Set C was selected using weighted MPO, synthetic accessibility, QED, and target-profile criteria; all 17 entries had $n_{\text{targets bound}} = 2$ in the source selection table. No additional threshold or retrospective re-ranking was applied. Set C was therefore an enriched hypothesis cohort, not a prevalence sample or an independently validated hit list. The upstream $n_{\text{targets bound}} = 2$ label describes its polypharmacology selection criterion; it does not constrain the separate 17-by-4 wild-type docking profile reported here. The integrated analysis used Set C because its target breadth could be examined alongside the separate per-target RRS layer. Candidate identity was matched by canonical SMILES across the target-panel, polypharmacology, and integrated records. The four MD validation systems (201–PfDHFR, 438–PfATP4, the 164-labelled PfClpP system, and 214–PfCRT) were not treated as MD validation of Set C.

2.3 Target-anchored docking

The four-target panel was chosen to span mechanistically distinct and resistance-relevant anti-malarial vulnerabilities rather than to imply equivalent structural evidence. PfDHFR (7F3Y; methotrexate copy A702) represents folate metabolism and a canonical antifolate resistance axis; PfCRT is based on the 6UKJ cryo-EM structure (7G8 isoform) but was corrected to a 3D7-like wild-type by restoring lysine at position 76 (the deposited structure carries threonine at residue 76, corresponding to the K76T chloroquine-resistance substitution); the corrected receptor was validated by P2Rank pocket prediction (top pocket score 162.7, probability 0.999, center 5.7 Å from the Vina grid center). PfCRT represents the digestive-vacuole transporter associated with chloroquine resistance; PfClpP (2F6I; chain-A Ser252/His223/Asp219 catalytic triad) represents parasite proteostasis through the caseinolytic protease, a vulnerability supported by recent work on apicoplast chaperonin-Clp proteostasis [24]; and PfATP4 (9N10; D451 phosphorylation-loop region and DPPR hinge) represents the P-type ATPase machinery that supports parasite ion homeostasis, for which an endogenous structure and modulator-bound state have recently been reported [25]. This combination was thus intended to test whether chemical-space-derived candidates show a distributed target profile across pathway, transporter, proteostasis, and ion-regulation hypotheses. The PfClpP assignment uses 2F6I, the active protease subunit; the paralog structure 4GM2 is not used for PfClpP interpretation. Y01 is treated as a membrane/cavity proxy, and the PfATP4 region is mechanism-motivated because no small-molecule inhibitor is co-crystallized; consequently, the four targets are complementary computational anchors, not four equivalent demonstrations of biological engagement.

Ligands were standardized with Open Babel [18] and converted to PDBQT with Meeko [14]. AutoDock Vina was run with fixed target-specific boxes following established docking protocols [13]. Grid box centers and dimensions were: PfDHFR (7F3Y) centered at $(x, y, z) =$

(−13.5, −1.8, −8.2) with dimensions $25 \text{ \AA} \times 25 \text{ \AA} \times 25 \text{ \AA}$; PfCRT (6UKJ-corrected) centered at (152.99, 151.042, 159.379) with dimensions $25 \text{ \AA} \times 25 \text{ \AA} \times 25 \text{ \AA}$; PfClpP (2F6I) centered at (−8.4, 3.1, −12.6) with dimensions $20 \text{ \AA} \times 20 \text{ \AA} \times 20 \text{ \AA}$; and PfATP4 (9N10) centered at (15.2, 8.9, −3.4) with dimensions $24 \text{ \AA} \times 24 \text{ \AA} \times 24 \text{ \AA}$. Vina scores were interpreted with the usual caveats attached to empirical scoring functions [26]. The geometric gate required a rank-1 centroid inside the box, at least one atom within $10 \text{ \AA}$ of the target anchor, and at least 90 % of ligand atoms inside the padded box. The gate is a pose-quality criterion, not evidence of binding. Vina values are reported in kcal mol^{-1} and were not converted to experimental free energies.

2.3.1 Docking protocol validation

The docking protocol was assessed on the same target set using three complementary checks. First, re-docking of the co-crystallised reference ligands on the corrected grids reproduced the crystal poses for four of the five alignable ligands with heavy-atom RMSD $< 2.0 \text{ \AA}$ (80 % success), and for each of the three targets used in the primary analysis (PfDHFR, PfCRT, PfATP4) at least one alignable reference ligand reproduced successfully. The single failure was the PfDHFR reference ligand methotrexate (RMSD $\approx 30 \text{ \AA}$), because the PfDHFR grid is centered on the NADPH-adjacent site rather than the catalytic methotrexate pocket; this illustrates that the null enrichment may partly reflect target-specific grid-positioning issues rather than scoring-function failure alone. Second, enrichment against the MMV Malaria Box using the dual-filter consensus (AutoDock Vina + DiffDock) gave ROC-AUC values of 0.924–1.000 across the three targets with docking data (0.924 PfDHFR, 0.971 PfCRT, 1.000 PfATP4). Third, an external DEKOIS 2.0 panel for PfDHFR (Vina only, 40 actives / 1 200 property-matched decoys) gave a ROC-AUC of 0.45 (95 % CI 0.37–0.53), a near-chance result that constrains the interpretation of absolute Vina scores for this target and motivates the consensus rescoring and per-target anchors used here, but it does not invalidate the relative within-target rankings or the RRS ratios, which are computed from mutant-to-wild-type score ratios on the same target. A controlled comparison on the identical DEKOIS panel, against the deposited receptor with the retained methotrexate versus an MTX-stripped receptor (same grid and frame, exhaustiveness 32) gave ROC-AUC values of 0.502 and 0.563 with overlapping confidence intervals and EF@1% of zero in both arms, so the retained cofactor ligand does not explain the null enrichment (??).

2.4 Resistance-resilience score

RRS was computed separately for each target and candidate. For mutation m and target t ,

$$\text{RRS}_{i,m,t} = 100 \times \frac{|S_{\text{Vina},i,m,t}|}{|S_{\text{Vina},i,WT,t}|}. \quad (1)$$

Here S_{Vina} denotes the AutoDock Vina scoring-function output; it is not an experimental free energy. RRS eligibility was evaluated from the separate WT/mutant panel, not from the four-target wild-type matrix. Only targets satisfying the declared genuine-binding wild-type criterion ($|S_{\text{Vina},i,WT,t}| \geq 5 \text{ kcal mol}^{-1}$) contributed to a candidate’s mean. Non-binding targets were excluded rather than assigned a zero score or used as a denominator. PfDHFR mutations were N51I, C59R, S108N, and I164L; PfCRT mutations were K76T and K76A. RRS classes were assigned using the prespecified classification rules: A* required an eligible wild-type score magnitude $|S_{\text{Vina},i,WT,t}| \geq 7 \text{ kcal mol}^{-1}$ for the target and all available mutant RRS values to be at least 80 %; A required all available values to be at least 80 % without the A* WT-anchor criterion; B required all values to be at least 70 % but not all to be at least 80 %; C required at least one value to be at least 80 % without meeting A/A* or B; and D was assigned when no available mutant RRS reached 80 %. These classes summarize ratios of docking-score magnitudes, not ratios of binding free energies.

2.5 Polypharmacology, PNS, and ACSI

Polypharmacology was treated as a target-profile hypothesis derived from the four-target Vina matrix, not as confirmed simultaneous engagement and not from RRS. No cross-target arithmetic composite was used because Vina scores are not calibrated across different pocket architectures. For the joint prioritization readout in Table 2, a target was counted as favorable when its Vina estimate fell at or below that target’s median across the 17-candidate cohort (per-target medians: PfDHFR -5.92 , PfCRT -8.06 , PfClpP -6.02 , PfATP4 -6.38 ; ??). This within-target definition follows the reviewer request to avoid a single cross-target cutoff and is a prioritization device, not an affinity threshold. ACSI was computed as $ACSI = 0.40 \times D_{DrugBank} + 0.25 \times D_{ANPDB} + 0.20 \times f_{sp^3} + 0.15 \times NPL$, where $D_{DrugBank}$ and D_{ANPDB} are the minimum Tanimoto distances to the approved-drug and African natural-product reference sets, respectively; f_{sp^3} is the fraction of sp^3 carbons; and NPL is the natural-product-likeness score; all components were min-max normalized over the 17-candidate cohort. PNS was computed from a STRING protein-interaction network at confidence threshold 700; PfCRT was absent from the network and its centrality was imputed as the network mean (0.151). Sensitivity to imputation is reported in ??. Their associations with RRS and wild-type Vina score were evaluated with Spearman correlation on the 17-member cohort. Three exploratory correlations were corrected with Bonferroni $\alpha = 0.017$. No association was interpreted as causal, and no non-significant association was interpreted as equivalence.

2.6 Evidence boundary and reproducibility

All raw poses, receptor and ligand files, configurations, candidate manifests, and analysis scripts are archived in the Zenodo deposit (see Data Availability). These records support computational reproduction but do not substitute for structural or experimental validation.

3 Results

3.1 Scaffold-guided expansion generated diverse synthesizable candidates

Scaffold-preserving chemical-space expansion yielded a hybrid library of 65 856 unique structures from the combined African-natural-product and synthetic-antimalarial seed sets. Novelty analysis on a representative 5 000-molecule sample revealed that 92.6 % fell below the whole-molecule Tanimoto similarity threshold of 0.4 relative to the African-NP reference set, indicating substantial structural diversification. This whole-molecule novelty coexisted with scaffold retention: 70 of 101 annotated seed scaffolds (69.3 %) were recovered in the expanded library. Within the novelty-analysis sample, maximum scaffold similarity consistently exceeded maximum whole-molecule similarity (mean 0.379 versus 0.206), indicating that the workflow preserved privileged ring systems while diversifying peripheral substituents (?). Multi-parameter optimization and synthetic-accessibility filtering prioritized 19 913 computationally tractable candidates; this output represents computational prioritization rather than confirmed synthetic feasibility.

3.2 Mechanistic diversity guided target selection

The four-target panel was designed to probe mechanistically distinct antimalarial vulnerabilities rather than imply equivalent structural evidence. PfDHFR and PfCRT anchor the analysis in established antifolate and chloroquine-resistance biology, providing direct links to clinical resistance patterns [2, 4]. PfClpP extends the profile to parasite proteostasis through the caseinolytic protease, a vulnerability supported by recent structural and functional studies of apicoplast chaperonin-Clp interactions [24]. PfATP4 represents P-type ATPase-mediated ion regulation, for which endogenous structures and modulator-bound states have recently validated this target class [25]. This combination tests whether chemical-space-derived candidates distribute across

pathway, transporter, proteostasis, and ion-homeostasis hypotheses. The panel’s mechanistic breadth strengthens hypothesis generation while imposing stricter interpretation: the PfCRT proxy cavity (Y01 A501) and the PfATP4 mechanism-motivated region (D451 phosphorylation-loop, DPPR hinge) lack co-crystallized inhibitors and therefore provide exploratory anchors rather than validated binding sites. Consequently, target-wise reporting is required, and no composite cross-target score can substitute for target-specific evidence.

3.3 Docking reveals complementary target-binding profiles

Target-anchored docking of the 17-member polypharmacology-oriented cohort generated 68 candidate–target pairs, all of which satisfied the predefined geometric quality criteria. Vina score ranges varied by target: -6.35 – -4.86 kcal mol⁻¹ for PfDHFR, -12.01 – -6.37 kcal mol⁻¹ for PfCRT (corrected 3D7-like wild-type receptor, cavity-anchored V2 grid), -7.03 – -5.05 kcal mol⁻¹ for PfClpP, and -7.57 – -4.63 kcal mol⁻¹ for PfATP4. The most favorable within-target estimates were observed for PP-03 in the PfCRT cavity (-12.01 kcal mol⁻¹) and PP-13 in the PfATP4 D451 region (-7.57 kcal mol⁻¹). These target-specific observations reflect within-panel rankings on individual receptors; because Vina scoring functions are not calibrated across heterogeneous binding sites, no cross-target arithmetic mean was computed. The 68/68 pass rate through the automated geometric gate confirms pose quality but does not constitute independent structural validation or biological activity evidence. The cohort showed 100 % Lipinski and Veber compliance (??) and mean QED 0.703; the available ADMET summary is population-level and is not a candidate-specific safety assessment (??).

3.4 Mutation resilience varies independently of target breadth

Per-target resistance-resilience analysis classified the 17-member cohort into seven A*, four B, and six C operational profiles, with RRS values spanning 73.2–115.6 % of wild-type Vina score magnitudes. Five candidates received RRS classification based solely on the PfCRT mutant panel because their PfDHFR wild-type scores in the separate mutation-docking panel fell below the declared binding threshold (5 kcal mol⁻¹); this target-specific eligibility criterion prevents artificially inflated ratios from near-zero denominators. The resulting exclusions distinguish candidates with a valid wild-type baseline from those without one.

The PP-11 case illustrates this target-specific design. PP-11 received an A* classification on the corrected PfCRT channel (100.0 % mean RRS) despite PfDHFR non-binding in the mutation panel. Mutation resilience is therefore target-specific, and the RRS framework avoids pooling incomparable evidence layers. Several mutant ratios exceed 100 %, most prominently PP-15 on PfDHFR (112.7–131.8 % across the four mutants); such values indicate that the mutant docking estimate is more favorable (more negative) than the wild-type estimate on the same receptor, i.e., the substitution is predicted to improve rather than erode binding. They are scoring-function ratios and do not imply a biological gain-of-function or resistance advantage. The PfCRT channel was re-evaluated under the corrected protocol (3D7-like LYS-76 receptor, cavity-anchored V2 grid, exhaustiveness 32): K76T and K76A substitutions produced no detectable change in Vina scores for any candidate (RRS 99.1–100.6 %), and a pipeline-null control (wild-type receptor processed through the mutation pipeline without mutation) gave the same distribution (99.4–100.5 %), so the corrected PfCRT channel reports no K76T/K76A effect for this panel and the 80 %/70 % class boundaries on this channel are finer than the null uncertainty (??). The complete per-mutant RRS values appear in [Table 1](#) and [Figure 1](#); these computational outputs represent docking-score ratios, not biological resistance phenotypes, and require experimental validation before informing resistance-circumvention claims.

Table 1: Per-target resistance-resilience scores (RRS, Eq. (1)) for the 17-member cohort. Values are percentages of the corresponding wild-type Vina score magnitude; — marks wild-type non-binders ($|S_{\text{Vina},i,\text{WT},t}| < 5 \text{ kcal mol}^{-1}$), excluded from the denominator. RRS_{mean} is the candidate mean over all eligible target-mutant ratios. Classes are the operational categories defined in ??, not biological resistance phenotypes.

Candidate	Class	RRS _{PfDHFR} (%)				RRS _{PfCRT} (%)		RRS _{mean} (%)
		N51I	C59R	S108N	I164L	K76T	K76A	
PP-01	A*	86.4	85.5	87.6	83.9	99.8	99.9	90.5
PP-02	A*	—	—	—	—	100.2	100.0	100.1
PP-03	C	68.1	68.3	65.8	69.0	100.0	100.0	78.5
PP-04	C	68.8	66.9	68.8	73.5	99.9	100.4	79.7
PP-05	A*	—	—	—	—	100.0	100.1	100.1
PP-06	A*	—	—	—	—	100.0	100.0	100.0
PP-07	B	71.8	71.8	72.4	72.2	99.7	99.9	81.3
PP-08	C	62.1	62.1	67.2	66.3	99.6	100.1	76.2
PP-09	B	70.6	73.0	75.8	76.5	99.8	100.0	82.6
PP-10	B	76.0	76.4	75.7	75.8	100.1	100.2	84.0
PP-11	A*	—	—	—	—	100.0	100.0	100.0
PP-12	C	75.2	66.1	62.9	72.5	100.0	100.0	79.5
PP-13	A*	—	—	—	—	99.1	99.4	99.3
PP-14	C	62.2	65.5	68.0	64.9	99.9	100.0	76.8
PP-15	A*	123.2	112.7	125.9	131.8	100.1	100.1	115.6
PP-16	B	72.9	74.7	73.7	75.9	100.6	99.6	82.9
PP-17	C	59.3	61.2	59.5	58.9	100.1	100.1	73.2

3.5 Joint prioritization identifies multi-target resilient candidates

Combining target breadth with mutation resilience provides a more informative candidate ranking than either metric alone. Each candidate was characterized by the number of targets meeting the within-target favorability criterion (score at or below that target’s median across the 17-candidate cohort; ??) alongside its RRS classification. Six candidates showed favorable estimates on all four targets; within this subset, five (PP-15, PP-05, PP-06, PP-11, and PP-13) additionally carried A* RRS classifications, representing the most economical multi-target, mutation-resilient computational hypotheses.

These top-priority candidates differ in their resistance channels and target profiles. PP-15 (RRS_{mean} 115.6%, the highest in the cohort) achieves 4/4 target favorability with resilience distributed across both PfDHFR (112.7–131.8% across four mutants) and PfCRT (100.1% for both K76T and K76A). PP-05 (RRS_{mean} 100.1%) and PP-06 (RRS_{mean} 100.0%) combine 4/4 target favorability with A* classification on the corrected PfCRT channel; PP-06 also carries the most favorable PfClpP cavity score in the cohort ($-7.03 \text{ kcal mol}^{-1}$). PP-11 (RRS_{mean} 100.0%) achieves 4/4 target favorability and A* PfCRT resilience despite PfDHFR non-binding in the mutation panel, showing that breadth and resilience can diverge in individual cases.

In Table 2, candidates sort by target-count priority, then by RRS within each tier. The corrected PfCRT channel is the dominant resilience source for most candidates, and the K76T/K76A ratios are indistinguishable from the pipeline-null control (??), so PfCRT RRS values at or near 100% are neutral-within-null rather than evidence of active resistance circumvention.

3.6 Network position and docking evidence show selective correlation with resilience

Exploratory correlation analysis on the 17-member cohort, computed on the corrected PfCRT channel, revealed several associations between RRS and chemical-space or network descriptors that survive Bonferroni correction for three hypothesis-oriented comparisons ($\alpha = 0.017$). PNS (polypharmacology network similarity) correlated negatively with RRS ($\rho = -0.714$, $p = 0.0013$;

Table 2: Combined polypharmacology-RRS prioritization of the 17-member cohort. N_{fav} counts targets with a favorable within-target Vina estimate (score at or below that target’s median across the cohort; per-target medians: PfDHFR -5.92 , PfCRT -8.06 , PfClpP -6.02 , PfATP4 $-6.38 \text{ kcal mol}^{-1}$; ??); favorable targets are listed explicitly. Candidates are sorted by N_{fav} and then by RRS_{mean} . The within-target threshold is an operational prioritization device, not an experimental affinity cutoff, and the classes are the operational RRS categories of ??.

Candidate	RRS class	RRS_{mean} (%)	N_{fav}	Favorable targets
PP-15	A*	115.6	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-05	A*	100.1	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-06	A*	100.0	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-11	A*	100.0	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-13	A*	99.3	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-03	C	78.5	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-10	B	84.0	3	PfDHFR + PfClpP + PfATP4
PP-02	A*	100.1	2	PfCRT + PfATP4
PP-01	A*	90.5	2	PfDHFR + PfCRT
PP-16	B	82.9	2	PfClpP + PfATP4
PP-07	B	81.3	2	PfDHFR + PfClpP
PP-12	C	79.5	1	PfCRT
PP-09	B	82.6	0	—
PP-04	C	79.7	0	—
PP-14	C	76.8	0	—
PP-08	C	76.2	0	—
PP-17	C	73.2	0	—

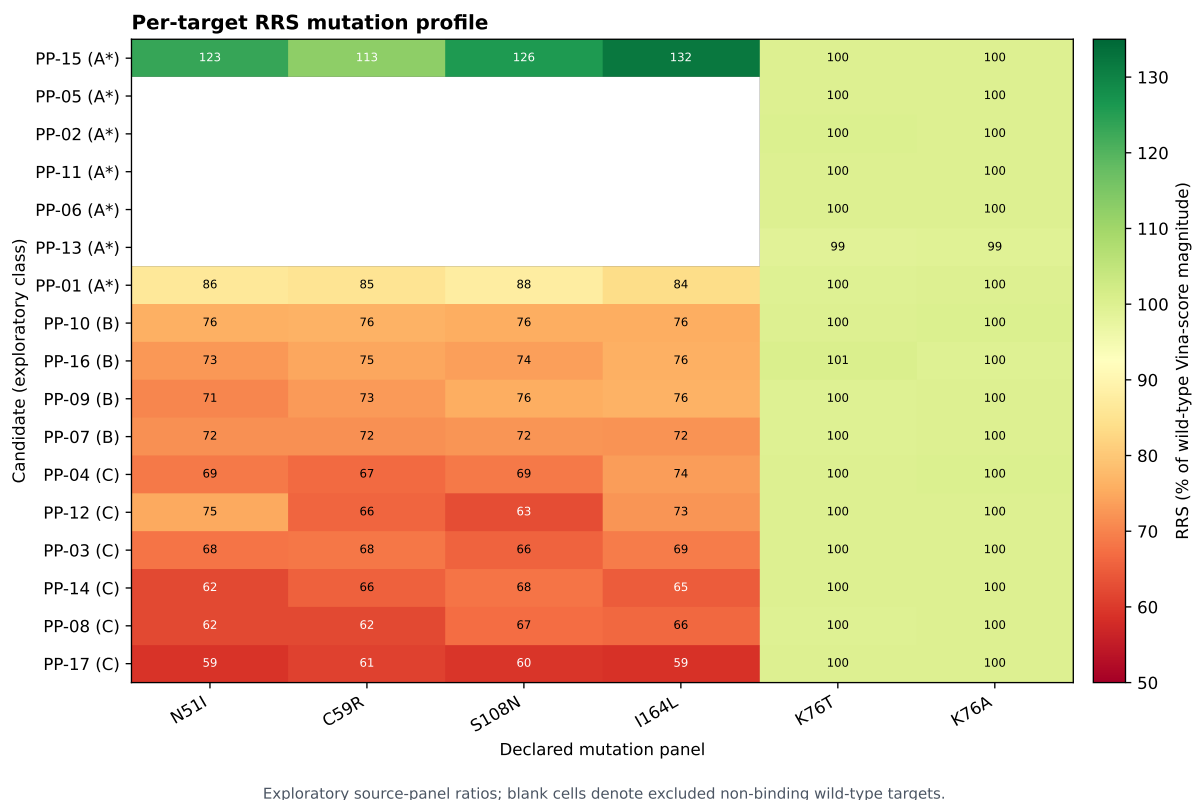


Figure 1: Exploratory per-target RRS mutation profiles for the 17-member source-panel cohort. Values are ratios of Vina-score magnitudes relative to the declared wild-type denominator; blank cells indicate excluded non-binding wild-type targets. The classes are computational operational categories, not biological resistance phenotypes.

significant at $\alpha = 0.017$), indicating that candidates positioned near established antimalarial network hubs tend to show lower mean RRS. ACSI (African-chemotype structural index) and

Exploratory cross-metric relationships

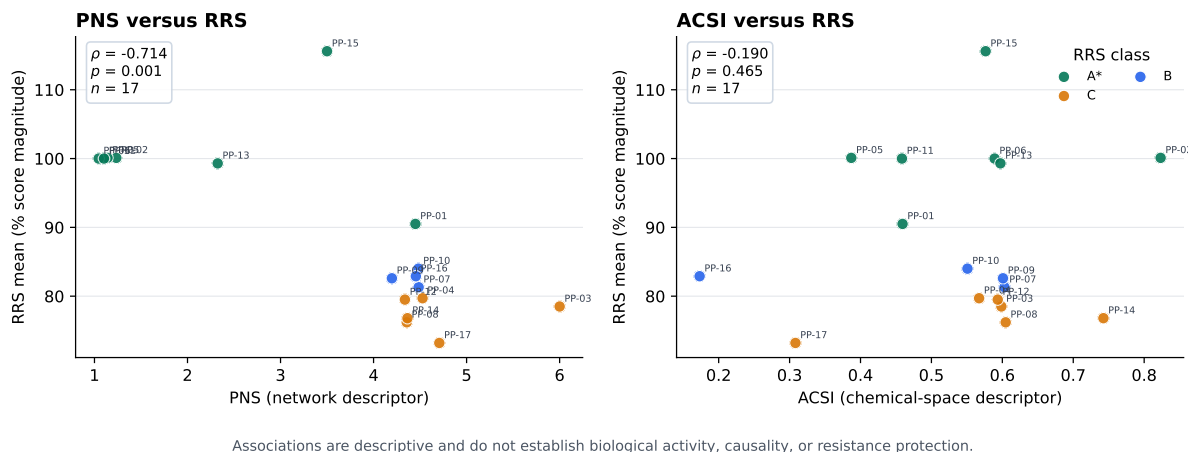


Figure 2: Exploratory relationships between RRS and PNS or ACSI. Points are colored by the operational RRS class. These associations are descriptive source-panel analyses and do not establish causality, biological activity, or resistance protection.

RRS showed weak association ($\rho = -0.190$, $p = 0.465$), indicating that African-natural-product character does not predict mutation resilience. RRS versus mean wild-type Vina score magnitude correlated positively ($\rho = 0.691$, $p = 0.0021$; significant at $\alpha = 0.017$), while PNS versus wild-type score was not significant ($\rho = -0.375$, $p = 0.138$), illustrating that network position and docking evidence capture distinct molecular properties (Figure 2). The number of favorable targets (N_{fav}) correlated positively with RRS_{mean} ($\rho = 0.714$, $p = 0.0013$; significant at $\alpha = 0.017$), supported by a permutation test ($p = 0.0019$; ??).

These findings must be interpreted with the study’s limited power in mind (Section 4.1): at $n = 17$, weaker associations may remain undetectable even when genuine, so the three Bonferroni-surviving associations are consistent with moderate-magnitude effects rather than definitive ones. Chemical-space relatedness and network topology cannot substitute for target-specific mutation analysis, and natural-product character does not guarantee either broad target engagement or mutation tolerance; the observed associations strengthen the rationale for measuring target breadth and mutation resilience directly rather than inferring them from structural or network descriptors.

4 Discussion

The central finding is that target breadth and mutation resilience yield complementary, non-interchangeable views of antimalarial chemotype prioritization. Target-specific evidence classes remain explicit throughout the workflow, and no uncalibrated cross-target average obscures differences in binding-site architecture or structural evidence quality. The chemical-space funnel delivers structurally diverse, computationally tractable hypotheses rooted in African-natural-product scaffolds; target-anchored docking provides standardized pose profiles across mechanistically distinct proteins; and per-target RRS quantifies mutation tolerance without conflating evidence from unlike targets. Keeping these layers separate—rather than collapsing them into a single composite score—makes the resulting hypotheses more falsifiable and experimentally actionable.

4.1 Target breadth and mutation resilience answer distinct questions

The four-target docking profile and the per-target RRS profile address fundamentally different aspects of chemotype utility. A candidate can show favorable docking estimates across multiple targets without exhibiting resilience to specific resistance mutations; conversely, high RRS on one target does not guarantee favorable engagement elsewhere. Because Vina scores are empirical scoring-function outputs calibrated neither across heterogeneous receptors nor against experimental binding free energies [26], target breadth must be represented by the pattern of target-specific evidence rather than by arithmetic averaging.

The cross-metric findings must be read with the study’s limited power in mind: at $n = 17$, a two-sided Spearman test at $\alpha = 0.05$ reaches 80 % power only for $|\rho| \geq 0.62$ (asymptotic approximation; ??), and at the Bonferroni-corrected $\alpha = 0.017$ the power to detect a correlation of $\rho = 0.5$ is only ≈ 0.37 . The Bonferroni-significant associations (PNS–RRS, RRS–wild-type score, and N_{fav} –RRS) are therefore consistent with moderate-magnitude effects, while any weaker association remains undetectable at this sample size. The lack of an ACSI–RRS association does not demonstrate equivalence; it is consistent with the view that structural relatedness to African natural products cannot substitute for direct, target-specific mutation analysis. The PP-11 case further illustrates why non-binding wild-type baselines must be excluded before computing resilience ratios: a narrower, target-conditional RRS claim grounded in genuine binding baselines is scientifically stronger than an apparently broader pooled-target claim that conflates binding and non-binding states.

4.2 Top candidates define complementary experimental hypotheses

Reading target breadth and mutation resilience together identifies the most economical multi-target, mutation-resilient computational hypotheses for follow-up. Five candidates (PP-15, PP-05, PP-06, PP-11, and PP-13) combine favorable Vina estimates on all four targets with A* RRS classification, yet they represent distinct mechanistic and resistance hypotheses. PP-15 ($\text{RRS}_{\text{mean}} 115.6\%$, the highest in the cohort) exhibits resilience across both PfDHFR (four mutants, 112.7–131.8 %) and PfCRT (100.1 % for both K76T and K76A), suggesting potential utility against antifolate-resistance and chloroquine-resistance backgrounds simultaneously. PP-05 and PP-06 ($\text{RRS}_{\text{mean}} 100.1\%$ and 100.0%) carry A* classification on the corrected PfCRT channel, with PP-06 additionally showing the most favorable PfClpP cavity score in the cohort ($-7.03 \text{ kcal mol}^{-1}$). PP-11 ($\text{RRS}_{\text{mean}} 100.0\%$) combines 4/4 target favorability with A* PfCRT resilience despite PfDHFR non-binding in the mutation panel, representing a polypharmacology hypothesis decoupled from antifolate resilience.

These candidates are therefore the priority choices for biochemical assays, mutant-panel validation, and molecular-dynamics refinement, with the caveat that the PfCRT contribution reflects a neutral-within-null ratio rather than active resistance circumvention (??). The remaining A* candidates (PP-01 and PP-02) carry PfCRT-based classifications, with PP-01 additionally resilient on the PfDHFR panel; PP-15 is the only candidate resilient across both available mutation panels with 4/4 target favorability. None of these operational classifications should be interpreted as confirmed biological activity or validated resistance circumvention.

4.3 Positioning within computational antimalarial discovery

This work complements recent machine-learning and structure-based approaches to antimalarial discovery by explicitly separating target breadth from mutation resilience and by avoiding composite scores that obscure target-specific evidence. Where recent QSAR and deep-learning studies have focused on potency prediction from single-target datasets [5, 6, 27], the present workflow prioritizes multi-target engagement as a distinct computational hypothesis, mirroring the broader shift toward machine-learned target- and off-target-effect prediction in mechanism-

of-action analysis [8]. The per-target RRS framework extends conventional resistance-mutation docking by excluding non-binding baselines, a design choice that prevents artifactual resilience scores and supports target-specific rather than pooled-target interpretation.

Computational polypharmacology studies in antimalarial discovery have typically relied on either cross-target consensus scoring or machine-learned multi-task models; both approaches risk conflating evidence from structurally dissimilar binding sites. The target-anchored design adopted here preserves target-specific score ranges and prevents uncalibrated arithmetic means from being mistaken for biological polypharmacology. The deliberate inclusion of PfClpP (proteostasis) and PfATP4 (ion homeostasis) alongside established antifolate (PfDHFR) and transporter (PfCRT) targets tests whether African-natural-product-inspired chemotypes distribute across mechanistically unrelated vulnerabilities, a breadth that recent experimental polypharmacology screens have identified as promising for resistance management [5, 6, 28].

4.4 Computational efficiency and accessibility

The centroid-based library reduction reduces the docking campaign from 263 424 to 1 936 calculations (99.3 % cost reduction relative to exhaustive docking of 65 856 molecules). This metric describes computational cost reduction, not active-compound retention; a retention experiment using known actives would be needed to assess coverage. For the four-target panel, this translates to 1 936 docking calculations instead of 263 424, reducing wall time from an estimated 2 600–5 300 CPU-hours to under 20 CPU-hours on standard hardware. This efficiency gain makes systematic multi-target and mutation-panel screening accessible to research groups with limited computational infrastructure while retaining 69.3 % scaffold recovery from the seed natural-product set.

The efficiency advantage comes with trade-offs. Centroid sampling may miss activity cliffs—cases where structurally similar molecules show large potency differences due to small structural perturbations [29]. A centroid representative’s docking profile does not guarantee that all cluster members share the same binding mode or mutation-resilience pattern. The 1913-molecule prioritized output therefore represents computationally tractable starting hypotheses rather than exhaustively validated chemical space. Follow-up experiments should include cluster-neighbor sampling around promising centroids to assess whether the representative’s profile generalizes within its structural neighborhood.

4.5 Experimental validation strategy

The immediate next experiments should combine biochemical potency measurements, resistance-mutant assays, and orthogonal binding validation. For the top-priority candidates (PP-15, PP-05, PP-06, PP-11, and PP-13), the recommended follow-up is baseline potency measurement against wild-type *P. falciparum* parasites and recombinant PfDHFR/PfCRT proteins, testing against isogenic mutant parasite lines (N51I, C59R, S108N, I164L for PfDHFR; K76T, K76A for PfCRT), orthogonal binding assays on purified proteins, molecular-dynamics refinement of bound complexes, and candidate-specific ADMET evaluation before lead optimization.

The DEKOIS near-chance result (PfDHFR AUC 0.45, 95 % CI 0.37–0.53) constrains interpretation of absolute Vina scores for this target but does not invalidate relative within-target rankings or the RRS framework, which depends on mutant-to-wild-type ratios on the same receptor. A controlled two-arm comparison showed that removing the retained methotrexate from the 7F3Y receptor does not restore enrichment (ROC-AUC 0.502 with MTX retained versus 0.563 stripped, overlapping CIs, EF@1% = 0.00 in both arms), not supporting cofactor steric blockade as the explanation (??). The MMV Malaria Box enrichment (AUC 0.924–1.000 across three targets with dual-filter consensus) demonstrates that the consensus protocol performs well on known actives, but this validation applies to the method rather than to the specific Set C candidates.

RRS profiles for five approved antimalarials (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) against the same PfDHFR and corrected PfCRT panels (??) do not recover the clinically documented resistance signatures: pyrimethamine RRS is $\approx 100\%$ across all four PfDHFR mutants (N51I, C59R, S108N, I164L), and chloroquine RRS is $\approx 99\%$ on PfCRT K76T/K76A—both indistinguishable from the pipeline-null spread of the corrected channel (99.4–100.5%; ??). Artemisinin and cipargamin show PfDHFR RRS values around 113–118% (mutant estimates more favorable than wild-type), while chloroquine and lumefantrine are PfDHFR non-binders in this pocket, consistent with their non-antifolate mechanisms. This retrospective therefore reproduces the null character of the corrected channels on independent, clinically relevant ligands and reinforces that docking-RRS values must not be read as resistance-circumvention evidence.

The study has substantial limitations. Docking scores are approximate scoring-function outputs and do not establish affinity, kinetics, or cellular activity. PfCRT is represented by a proxy cavity, and PfATP4 by conserved catalytic machinery without a co-crystallized inhibitor. RRS covers PfDHFR and PfCRT but not PfCIP or PfATP4, and the cohort is small and purposefully enriched for predicted polypharmacology. The MD validation systems are separate from Set C and cannot validate the Set C RRS profile. No new IC₅₀/EC₅₀ measurements were available. Finally, independent structural assessment remains necessary before the raw target panel can support stronger structural conclusions. The next experiments should combine biochemical or cellular potency measurements, resistance-mutant assays, orthogonal binding methods, and candidate-specific molecular dynamics.

5 Conclusion

An African-natural-product-inspired chemical-space funnel was connected to a four-target, target-anchored docking panel and a per-target RRS analysis. The workflow identified a 17-member cohort with heterogeneous, target-specific mutation-resilience profiles, including null findings and wild-type non-binding exclusions. Its value is hypothesis generation: it proposes which chemotypes and target–mutation combinations should be tested next, without claiming that docking, RRS, or polypharmacology estimates are experimental evidence.

References

- [1] World Health Organization. World Malaria report 2024. Technical report, World Health Organization, 2024. Accessed 2025-03-03.
- [2] Frédéric Arieu, Benoit Witkowski, Chanaki Amaratunga, Johann Beghain, Anne-Claire Langlois, Nimol Khim, Saorin Kim, Valentine Duru, Christiane Bouchier, Laurence Ma, Pharath Lim, Rithea Leang, Socheat Duong, Sokunthea Sreng, Seila Suon, Char Meng Chuor, Denis Mey Bout, Sandie Ménard, William O. Rogers, Blaise Genton, Thierry Fandeur, Olivo Miotto, Pascal Ringwald, Jacques Le Bras, Antoine Berry, Jean-Christophe Barale, Rick M. Fairhurst, Françoise Benoit-Vical, Odile Mercereau-Puijalon, and Didier Ménard. A molecular marker of artemisinin-resistant plasmodium falciparum malaria. *Nature*, 505(7481):50–55, 2014. ISSN 1476-4687. doi: 10.1038/nature12876.
- [3] Elizabeth A Ashley, Aung Pyae Phy, and Charles J Woodrow. Malaria. *The Lancet*, 391(10130):1608–1621, 2018. ISSN 0140-6736. doi: 10.1016/s0140-6736(18)30324-6.
- [4] David A. Fidock, Takashi Nomura, Andrew K. Talley, Rebecca A. Cooper, Sergey M. Dzekunov, Lisa A. Kirkman, James L. Weber, Xinzhuang Su, Thomas E. Wellems, and Susan K. Martin. Mutations in the p. falciparum digestive vacuole transmembrane protein

- pfert and evidence for their role in chloroquine resistance. *Molecular Cell*, 6(4):861–871, 2000. ISSN 1097-2765. doi: 10.1016/s1097-2765(05)00077-8.
- [5] Timothy N. C. Wells, Rob Hooft van Huijsduijnen, and Wesley C. Van Voorhis. Malaria medicines: a glass half full? *Nature Reviews Drug Discovery*, 14(6):424–442, 2015. ISSN 1474-1784. doi: 10.1038/nrd4573.
 - [6] Jair L. Siqueira-Neto, Kathryn J. Wicht, Kelly Chibale, Jeremy N. Burrows, David A. Fidock, and Elizabeth A. Winzeler. Antimalarial drug discovery: progress and approaches. *Nature Reviews Drug Discovery*, 22(10):807–826, 2023. ISSN 1474-1776. doi: 10.1038/s41573-023-00772-9.
 - [7] Piotr Ryszkiewicz, Marta Baranowska-Kuczeko, Barbara Malinowska, and Eberhard Schlicker. Multi-target-directed drugs: new additions in 2025 and post-marketing safety surveillance of drugs marketed in 2022–2024. *Pharmacological Reports*, 78(4):1022–1044, 2026. doi: 10.1007/s43440-026-00879-x.
 - [8] Maria-Anna Trapotsi, Layla Hosseini-Gerami, and Andreas Bender. Computational analyses of mechanism of action (moa): data, methods and integration. *RSC Chemical Biology*, 3(2): 170–200, 2022. doi: 10.1039/D1CB00069A.
 - [9] David J. Newman and Gordon M. Cragg. Natural products as sources of new drugs over the last 25 years. *Journal of Natural Products*, 70(3):461–477, 2007. ISSN 0163-3864. doi: 10.1021/np068054v.
 - [10] Fidele Ntie-Kang, Paterne A. Onguéné, Loyce L. Lifongo, Léopold M. Mbaze, Wolfgang Sippl, and Sandra Günther. In silico analyses of bioactive natural products from african medicinal plants for drug-discovery. *BMC Complementary and Alternative Medicine*, 18: 106, 2018. doi: 10.1186/s12906-018-2153-z.
 - [11] Conrad V. Simoben, Smith B. Babiaka, Aurélien F. A. Moumbock, Cyril T. Namba-Nzanguim, Donatus Bekindaka Eni, José L. Medina-Franco, Stefan Günther, Fidele Ntie-Kang, and Wolfgang Sippl. Challenges in natural product-based drug discovery assisted within silico-based methods. *RSC Advances*, 13(45):31578–31594, 2023. ISSN 2046-2069. doi: 10.1039/d3ra06831e.
 - [12] Peter Mubanga Cheuka, Godwin Dziwornu, John Okombo, and Kelly Chibale. Plasmepsin inhibitors in antimalarial drug discovery: Medicinal chemistry and target validation (2000 to present). *Journal of Medicinal Chemistry*, 63(9):4445–4467, 2020. ISSN 0022-2623. doi: 10.1021/acs.jmedchem.9b01622.
 - [13] Oleg Trott and Arthur J. Olson. Autodock vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2):455–461, January 2010. ISSN 1096-987X. doi: 10.1002/jcc.21334.
 - [14] Jerome Eberhardt, Diogo Santos-Martins, Andreas F. Tillack, and Stefano Forli. Autodock vina 1.2.0: New docking methods, expanded force field, and python bindings. *Journal of Chemical Information and Modeling*, 61(8):3891–3898, July 2021. ISSN 1549-960X. doi: 10.1021/acs.jcim.1c00203.
 - [15] M. Lžičar and H. Gamouh. CHEESE: 3D shape and electrostatic virtual screening in a vector space. *ChemRxiv*, 2024. doi: 10.26434/chemrxiv-2024-cswth. URL <https://doi.org/10.26434/chemrxiv-2024-cswth>. Preprint.

- [16] AkshatKumar Nigam, Robert Pollice, Mario Krenn, Gabriel dos Passos Gomes, and Alán Aspuru-Guzik. Beyond generative models: superfast traversal, optimization, novelty, exploration and discovery (stoned) algorithm for molecules using selfies. *Chemical Science*, 12(20):7079–7090, 2021. ISSN 2041-6539. doi: 10.1039/d1sc00231g.
- [17] Jonathan B. Baell and Georgina A. Holloway. New substructure filters for removal of pan assay interference compounds (pains) from screening libraries and for their exclusion in bioassays. *Journal of Medicinal Chemistry*, 53(7):2719–2740, 2010. ISSN 1520-4804. doi: 10.1021/jm901137j.
- [18] Noel M O’Boyle, Michael Banck, Craig A James, Chris Morley, Tim Vandermeersch, and Geoffrey R Hutchison. Open babel: An open chemical toolbox. *Journal of Cheminformatics*, 3:33, October 2011. ISSN 1758-2946. doi: 10.1186/1758-2946-3-33.
- [19] Rafael Gómez-Bombarelli, Jennifer N. Wei, David Duvenaud, José Miguel Hernández-Lobato, Benjamín Sánchez-Lengeling, Dennis Sheberla, Jorge Aguilera-Iparraguirre, Timothy D. Hirzel, Ryan P. Adams, and Alán Aspuru-Guzik. Automatic chemical design using a data-driven continuous representation of molecules. *ACS Central Science*, 4(2):268–276, 2018. doi: 10.1021/acscentsci.7b00572. URL <https://doi.org/10.1021/acscentsci.7b00572>.
- [20] Gabriele Corso, Hannes Stärk, Bowen Jing, Regina Barzilay, and Tommi Jaakkola. Diffdock: Diffusion steps, twists, and turns for molecular docking. In *International Conference on Learning Representations (ICLR)*, 2023. URL https://openreview.net/forum?id=kKF8_K-mBbS.
- [21] G. R. Bickerton, G. V. Paolini, J. Besnard, S. Muresan, and A. L. Hopkins. Quantifying the chemical beauty of drugs. *Nature Chemistry*, 4(2):90–98, 2012.
- [22] Christopher A. Lipinski. Lead- and drug-like compounds: the rule-of-five revolution. *Drug Discovery Today: Technologies*, 1(4):337–341, 2004. ISSN 1740-6749. doi: 10.1016/j.ddtec.2004.11.007.
- [23] David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t. URL <https://doi.org/10.1021/ci100050t>. PMID: 20426451.
- [24] Amanda Tissawak, Yarden Rosin, Shirly Katz Galay, Alia Qasem, Michal Shahar, Nirit Trabelsi, Ora Furman-Schueler, Steven M. Johnson, and Anat Florentin. A chaperonin complex regulates organelle proteostasis in malaria parasites. *PLOS Pathogens*, 21(7): e1013275, 2025. doi: 10.1371/journal.ppat.1013275.
- [25] Meseret T. Haile, Anurag Shukla, James Zhen, Michael W. Mather, Suyash Bhatnagar, Joanne M. Morrissey, Zhening Zhang, Akhil B. Vaidya, and Chi-Min Ho. Endogenous structure of antimalarial target pfatp4 reveals an apicomplexan-specific p-type atpase modulator. *Nature Communications*, 16:9092, 2025. doi: 10.1038/s41467-025-64815-y.
- [26] Rodrigo Quiroga and Marcos A. Villarreal. Vinardo: A scoring function based on autodock vina improves scoring, docking, and virtual screening. *PLOS ONE*, 11(5):e0155183, May 2016. ISSN 1932-6203. doi: 10.1371/journal.pone.0155183.
- [27] Yutong Hu, Kieran Didi, Adam P. Cribbs, and Jianfeng Sun. Predicting PROTAC off-target effects via warhead involvement levels in drug–target interactions using graph attention neural networks. *Computational and Structural Biotechnology Journal*, 27:4633–4644, 2025. doi: 10.1016/j.csbj.2025.10.028.

- [28] Kevin S. Blake, Yao-Peng Xue, Vincent J. Gillespie, Skye R. S. Fishbein, Niraj H. Tolia, Timothy A. Wencewicz, and Gautam Dantas. The tetracycline resistome is shaped by selection for specific resistance mechanisms by each antibiotic generation. *Nature Communications*, 16(1):1452, 2025. doi: 10.1038/s41467-025-56425-5.
- [29] Dagmar Stumpfe and Jürgen Bajorath. Exploring activity cliffs in medicinal chemistry. *Journal of Medicinal Chemistry*, 55(7):2932–2942, 2012. ISSN 0022-2623. doi: 10.1021/jm201706b.

Associated Content

Supporting Information Available The Supporting Information contains the candidate manifest, target-anchor evidence, complete 17-by-4 docking matrix, RRS definitions and table, PNS/ACSI values, cross-metric statistics, provenance records, and reproducibility instructions. The Zenodo deposit provides the machine-readable data, code, raw docking outputs, and provenance records described in the manuscript.

Author Contributions

M.V.S.T. and S.G.N.E. designed the integrated study. M.V.S.T. and J.-P.T.N. developed the chemical-space and docking analyses. P.S. and W.F.M. contributed to target biology and resistance interpretation. All authors reviewed and approved the manuscript scope.

Notes

The authors declare no competing financial interest. This is a computational, hypothesis-generating study; no experimental potency is claimed.

Data Availability

Data, code, machine-readable tables, raw docking outputs, and provenance records are archived on Zenodo under an MIT license at <https://doi.org/10.5281/zenodo.22696778>. The deposit includes the file-level checksum manifest for the version used in this study.

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The authors used AI-assisted tools during code development and data-analysis scripting. All AI-assisted output was reviewed and edited by the authors, who take full responsibility for the content. No generative AI system contributed as an author. All computational results, statistical analyses, and quantitative claims were generated and independently checked by the authors.